Brain Signal Processing: where we are

Detecting brain and mental disorders from resting-state EEG with graph neural networks · team update and full work history · 13 November 2026 · supervisor Dr. Nileshchandra Pikle

In one paragraph. We built and honestly tested about 30 models on 12 public EEG datasets. Our best scores are strong for Alzheimer's (AUC 0.92) and depression (0.97 on one dataset), decent for FTD (0.85) and schizophrenia (0.68-0.83 depending on hospital), and weak for telling Alzheimer's apart from FTD (0.68-0.86). The main lesson: **a bigger or fancier model does not help** - simple logistic regression ties our graph networks and transformers. What limits us is the data: small datasets, each hospital leaving a strong 'fingerprint' in its recordings, noisy labels in one dataset, and a recording mismatch in another. We also showed that several published high scores come from flawed testing.

0.92	0.85	0.97	0.68-0.83	0.68 / 0.86
Alzheimer's vs healthy (Thessaloniki)	FTD vs healthy (Thessaloniki)	Depression vs healthy (Mumtaz)	Schizophrenia vs healthy (3 hospitals)	Alzheimer's vs FTD (Thessaloniki / BrainLat)

Numbers are AUC (see 'How to read the numbers'). Each comes from people the model never saw during training.

How to read the numbers

Term	Plain meaning
AUC	Chance that the model ranks a random patient as 'more ill' than a random healthy person. 0.5 = coin flip, 0.7 = useful, 0.8 = good, 0.9+ = very good.
Subject-wise testing	A person's recording is either all in training or all in testing, never split. Splitting one person's 10-second windows across both (common in papers) inflates scores, because the model just recognises the person.
Leakage	Any way the test answers sneak into training or model choice, e.g. picking the best training epoch by looking at test accuracy. Gives scores that do not hold up.
Hospital effect	Each hospital's equipment and procedure leave a fingerprint. Our models could tell the hospital from healthy people alone with AUC 0.88-1.00 - easier than the disease.
Ensemble	Several models vote (we average their rankings). Members are fixed before seeing results.
Master model	One model per disease trained on people from all hospitals mixed together, tested on held-out people from every hospital.
95% CI	Range the true score probably lies in. With 30-100 people per dataset these ranges are wide (often ±0.10), so small differences between models mean nothing.

Current scope

We narrowed to the four disorders with the best and most defensible data: **Alzheimer's (AD), frontotemporal dementia (FTD), depression and schizophrenia**. Parkinson's was explored (4 datasets, AUC 0.73-0.80) and parked to keep focus.

Disorder	Datasets used	People	Notes
AD, FTD	Thessaloniki (OpenNeuro ds004504)	36 AD, 23 FTD, 29 healthy	Clean; one hospital
	BrainLat (Argentina + Chile, Synapse)	AD, FTD, healthy (86 usable)	Patients and healthy people were recorded into different file types - see caveats
Depression	Mumtaz (Malaysia, figshare)	30 depressed, 28 healthy	Clinical diagnosis; now our only depression set
Schizophrenia	ASEEG, Warsaw, Semmelweis	101 / 28 / 71	Three hospitals

Dropped datasets

Dataset	Why dropped
UNM depression (ds003478)	'Depressed' meant a questionnaire score ≥13. Of 46 such students only 11 had depression confirmed by clinical interview; 9 were interviewed and did not meet criteria. Label noise capped scores at ~0.6-0.7 whatever we did.
ASZED schizophrenia	Eyes-open and only ~20 s per person (other sets are eyes-closed, minutes long).
14 Chilean BrainLat controls	Their signal files do not exist on Synapse (header files only).
Parkinson's sets (4)	Out of focus, kept in the repo.

Shared processing pipeline

Every recording goes through the same steps: 19 standard scalp electrodes → mains-noise filter → resample to 128 Hz → keep 1-45 Hz → average reference → cut into 10-second windows → throw away windows with extreme or flat signal → up to 30 clean windows per person. From each window we compute power in 6 frequency bands per electrode and how strongly each pair of electrodes moves together (coherence). These become the nodes and edges of the brain graph.

Work history, in order

1. Proposal and baseline

- Rejected the suggested RL-based paper (IEEE 10977776) as a baseline - not a GNN-on-EEG method.
- Reproduced **EEG-GCNN** (released checkpoints, 478 held-out people) as our reference GNN.

2. Literature plan

- Reviewed 6 papers. Chose **MSBP-GNN** as the main model to rebuild, **Tigga et al.** as an example of flawed testing, and the GATv2→LSTM idea (spatio-temporal) as an extension.

3. Data and first models

- Built the raw-EEG pipeline and collected 12 public datasets.
- Trained GATv2 and GCN per disorder, rebuilt MSBP-GNN, built GATv2-LSTM. All landed in the same range; the LSTM added nothing.

4. Cross-hospital attempt (paused)

- Tried three standard fixes for hospital differences: ComBat, Euclidean alignment, domain-adversarial training (DANN). Transfer to an unseen hospital stayed near chance (e.g. schizophrenia).
- Site probe: hospital identifiable at AUC 0.88-1.00 from healthy people alone. Paused, then dropped.

5. Pushing the scores up honestly

• Added richer EEG markers (1/f slope, alpha peak, entropy, Hjorth), a Riemannian model, the pretrained LaBraM transformer, and more datasets.

• Built **Ensemble v2** (6 models, fixed in advance) - our headline within-dataset numbers.

• Target '0.85 everywhere' was not reachable honestly; instead we added clinically sharper secondary tasks (e.g. moderate Parkinson's 0.96).

6. Teammate plan 1 (rigorous AD/FTD experiments)

• **3-class AD / FTD / healthy**: accuracy 0.65, macro-F1 0.62 - FTD is mostly confused with AD.

• **DT-GCN audit** (paper claims 82%): their code picks the best epoch on the test fold. Same model with honest selection: 0.52 accuracy; strict last-epoch: 0.38.

• **Graph ablation** (6 graph types × GCN/GraphSAGE/GATv2): a model with **no edges** scores the same as every graph (F1 0.48-0.55). Edges add nothing measurable.

• Re-ran Tigga et al.: edges inert; their leaky protocol 0.48 vs honest 0.44.

7. Focus on top 4 and BrainLat

• Narrowed to AD, FTD, depression, schizophrenia. Downloaded BrainLat (Synapse).

• Found that BrainLat patients and healthy people come from different file types (4× amplitude gap) - patient-vs-healthy scores of 1.00 were fake. Reviewed how others used BrainLat: careful studies get ~0.68-0.70.

• AD vs FTD on BrainLat is clean (same file type): 0.74 within BrainLat; 0.70 / 0.66 when trained on one dataset and tested on the other.

8. Teammate plan 2 (transformers, master model)

• Built the 'new' components from the teammate table: Graph Transformer, attention-MIL, GroupDRO, LaBraM fine-tuning - all trained as one **master model per disease** on all hospitals mixed.

• Result: Graph Transformer ties the simple models, LaBraM fine-tuning is mixed, attention-MIL and GroupDRO are worse. Weighting hospitals equally changed nothing. **AD vs FTD improved clearly** with the master model.

• Dropped leave-one-hospital-out and UNM.

9. Preprocessing review (running)

• Literature: ICA / ASR artifact removal and reference changes do not improve classification (one study on our own Thessaloniki data). Testing three cheap changes now: windows spread across the whole recording, 1/f-corrected band power, and wPLI connectivity.

What worked and what did not

What we tried	Result	Verdict
Ensemble v2 (6 models voting)	Best within-dataset scores	Worked
Logistic regression / Riemannian / rich markers	Match or beat the graph networks	Worked
Master model per disease (all hospitals mixed)	Big gain for AD vs FTD; small losses elsewhere	Partly
Honest audits of published models (DT-GCN, Tigga)	Showed their scores rely on leaky testing	Worked
GATv2 / GCN / MSBP-GNN	All similar, none beats simple models	Partly
Graph edges (6 graph types)	No better than no edges	Did not
GATv2-LSTM (time sequence)	No gain	Did not
Graph Transformer	Ties the simple models	Partly
LaBraM pretrained transformer (frozen / fine-tuned)	Useful ensemble member, not a winner	Partly
Attention-MIL, GroupDRO	Worse on every task	Did not

What we tried	Result	Verdict
ComBat, Euclidean alignment, DANN (hospital fixes)	Unseen hospital still near chance	Did not
Hospital-balanced training	No consistent change	Did not
Stricter depression cut-off (UNM)	No improvement; dataset dropped	Did not
3-class AD / FTD / healthy	65% accuracy; FTD confused with AD	Partly
Preprocessing v2	Results pending	Running

Why the fancy models do not win

- **Too few people.** 28-130 people per dataset. A transformer with thousands of parameters memorises individuals; a simple model cannot.
- **Hospital fingerprints are stronger than disease.** Any flexible model happily learns the hospital.
- **The information is mostly in band power** (how much slow vs fast activity). Graph structure and time order add little on top - the ablation proved it for edges.

Best results per disorder

Task	Within one dataset (Ensemble v2)	Master model, all hospitals (ensemble)
AD vs healthy	Thessaloniki **0.92**	Thessaloniki 0.91 · BrainLat 0.74*
FTD vs healthy	Thessaloniki **0.85**	Thessaloniki 0.81 · BrainLat 0.65*
AD vs FTD	Thessaloniki 0.58 · BrainLat 0.74	**Thessaloniki 0.68 · BrainLat 0.86**
Depression	Mumtaz **0.97**	- (single dataset now)
Schizophrenia	ASEEG 0.83 · Warsaw 0.70 · Semmelweis 0.68	0.81 · 0.67 · 0.67
3-class AD/FTD/healthy	accuracy 0.65 · macro-F1 0.62	not run yet

* BrainLat patient-vs-healthy carries the file-type caveat. Within-dataset and master numbers use different splits, so compare them loosely.

Master model, every model (AUC per hospital)

Results file: results/pooled_deep.json. All models used identical splits; differences under ~0.10 are within noise.

AD vs control

Model	Thessaloniki	BrainLat	All
Logistic regression	0.92	0.73	0.83
GATv2 (graph)	0.89	0.67	0.79
Graph Transformer	0.89	0.74	0.82
Attention-MIL	0.82	0.64	0.74
Attention-MIL + GroupDRO	0.83	0.69	0.76
LaBraM fine-tuned	0.88	0.72	0.81
Ensemble (all 6)	0.91	0.74	0.83

FTD vs control

Model	Thessaloniki	BrainLat	All
Logistic regression	0.76	0.69	0.72
GATv2 (graph)	0.79	0.69	0.73
Graph Transformer	0.81	0.58	0.70
Attention-MIL	0.67	0.62	0.65
Attention-MIL + GroupDRO	0.73	0.60	0.67
LaBraM fine-tuned	0.73	0.52	0.64
Ensemble (all 6)	0.81	0.65	0.73

AD vs FTD

Model	Thessaloniki	BrainLat	All
Logistic regression	0.62	0.86	0.74
GATv2 (graph)	0.67	0.79	0.72
Graph Transformer	0.69	0.83	0.75
Attention-MIL	0.51	0.66	0.58
Attention-MIL + GroupDRO	0.57	0.71	0.64
LaBraM fine-tuned	0.62	0.68	0.65
Ensemble (all 6)	0.68	0.86	0.76

Schizophrenia

Model	ASEEG	Warsaw	Semmelweis	All
Logistic regression	0.75	0.72	0.69	0.72
GATv2 (graph)	0.81	0.79	0.63	0.75
Graph Transformer	0.78	0.70	0.56	0.70
Attention-MIL	0.73	0.63	0.60	0.68
Attention-MIL + GroupDRO	0.70	0.52	0.61	0.65
LaBraM fine-tuned	0.79	0.44	0.70	0.72
Ensemble (all 6)	0.81	0.67	0.67	0.74

Status of the teammate's architecture table

Component	Status	Verdict
GATv2, GCN, MSBP-GNN, GATv2+LSTM	Built and benchmarked	Done
Riemannian, LaBraM, ComBat, Euclidean alignment, DANN	Built and benchmarked	Done
Graph Transformer	Built (master model) - ties simple models	Done
LaBraM deeper fine-tuning	Built (last 2 blocks) - mixed	Done
Attention-based MIL	Built - worse	Done
GroupDRO	Built - worse	Done
Uncertainty-aware prediction	Next step (calibration + 'unsure' output)	Next
Spectral / hierarchical temporal transformer, cross-attention	Skipped: need far more data; LaBraM covers the idea	Dropped
Supervised contrastive learning	Skipped for now: little to contrast with ~100 people	Dropped

Honest caveats to keep in the report

- **Depression is single-site** (Mumtaz, 58 people). 0.97 cannot be claimed to generalise. Option: request access to MODMA (clinically diagnosed depression, needs a signed request).
- **BrainLat patient-vs-healthy** is confounded by recording file type. We remove the amplitude gap, but other differences may remain. AD vs FTD on BrainLat is clean.
- **Small datasets** mean wide confidence intervals; many model differences are noise.
- **Generalising to a brand-new hospital is not solved** - we tried and documented it.

What we do next

#	Task	How	Time
1	Finish preprocessing v2	Running now. Adopt only if it helps consistently beyond the noise.	~1 h
2	Freeze the final system	Master model for AD, FTD, AD-vs-FTD, schizophrenia; within-dataset for depression. Members fixed - no more picking by results.	~30 min
3	Calibrated probabilities + 'unsure'	Per disease, fitted on training data only. Curves like 'answers 70% of people at X% accuracy'.	1-2 h
4	Explainability	Which electrodes and frequency bands drive each disease, checked for stability across folds.	~1 h
5	Update report, results page, model registry	Add everything above.	1-2 h

#	Task	How	Time
6	GitHub	Push branch gcn-comparison and open a PR to master (after team sign-off).	5 min
7	Optional: 3-class with BrainLat	Same 3-class test on the master model.	~30 min

Decisions for the team

- Is single-site depression acceptable, or do we request MODMA?
- Agree the headline framing: an honest benchmark showing that simple models match GNNs/transformers on current public EEG, plus audits of leaky published results - rather than chasing one big number.

Code: github.com/AbdulMalik1287/MiniProject-I, branch gcn-comparison (latest commits local until pushed). Key files: pooled_deep.py (master models), ensemble.py, three_class.py, dtgcn.py, graph_ablation.py, prep_v2.py, docs/MODEL_REGISTRY.md.